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Your agent needs a sandbox, not a desert

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whoami - Samuel Colvin

- Creator of **Pydantic**
- Founder & CEO of **Pydantic Labs**

The Pydantic Stack

- Validation
- Agent Framework
- Observability - AI & Infra
- Evals
- Prompt management

One platform. One trace.

**Replace your separate infra and AI
observability platforms with
Pydantic Logfire.**

Agents & Tools

Agents are easy to define in the abstract.

```
env = Environment()
tools = Tools(env)
system_prompt = "Goals, constraints, and how to act"

while True:
    action = llm.run(system_prompt + env.state)
    env.state = tools.run(action)
```

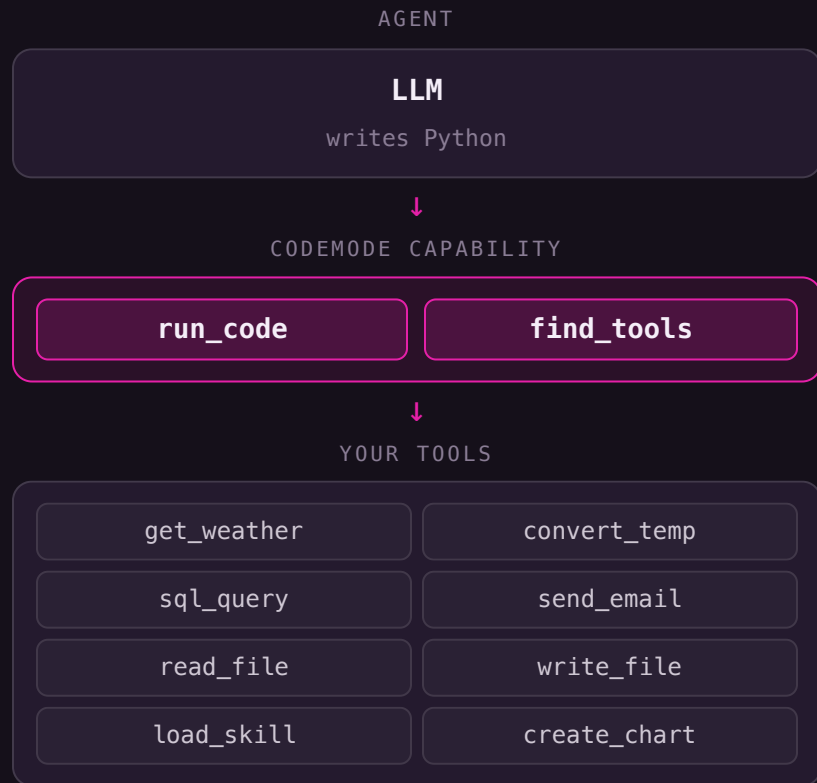
Agents are models using tools in a loop

... but how many tools?

```
{
    "get_weather",
    "search_flights",
    "book_hotel"
}
```

Enter Codemode 🥁

- aka "Programatic tool calling", CodeAct, and RLM (Recursive Language Models)
- The model writes Python (or js); the Python calls tools
- Tool composition, branching, looping - all happen in **code**, not round-trips
- Intermediate state stays in the runtime, not in the context window
- The agent only ever sees **two** tools: `run_code` and `find_tools`
- "1000 tools in 1000 tokens" is the promise



But how do we run it?

Two Paradigms

Both alike in dignity.

Light tasks

- Calculations
- Run queries
- Pass tool outputs to next tool
- Render a chart

TYPICAL DURATION

milliseconds

Heavy tasks

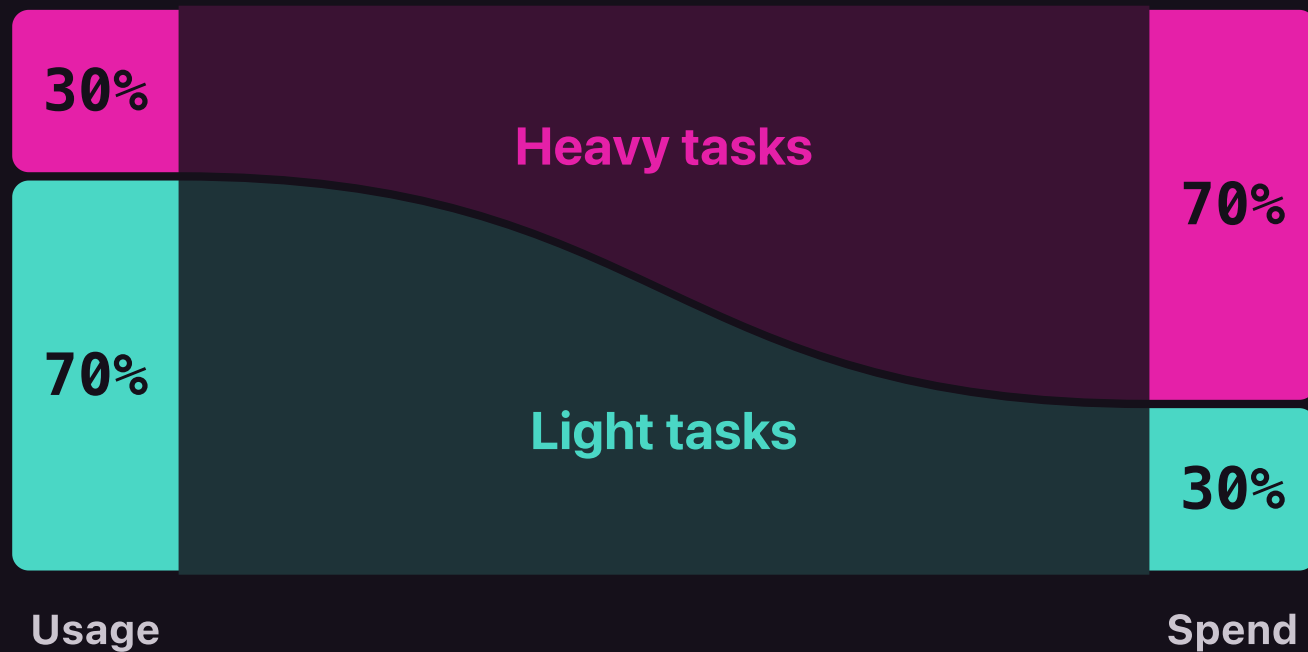
- Clone a repo
- Run a coding agent
- Heavy computation
- Compilation

TYPICAL DURATION

minutes

Two Paradigms

Both alike in dignity. **Usage**



Two Paradigms

Both alike in dignity. **Problems**

Light tasks

Execution environments:

- VM
- Container
- gVisor
- Firecracker

TYPICAL BOOT TIME

1–5s

Heavy tasks

Execution environments:

- VM
- Container
- gVisor
- Firecracker

TYPICAL BOOT TIME

1–5s

The sandbox vs. the desert

THE SANDBOX

sql_query

create_chart

create_table

load_skill

THE DESERT

awk sed gcc g++ clang less more
grep egrep fgrep make cmake
bash zsh sh fish curl wget
tar gzip bzip2 xz zip unzip
find xargs locate vim nano
emacs ed python3 pip pipx node
npm pnpm yarn git svn hg ssh
scp rsync cron systemd apt
apt-get dpkg yum dnf brew jq yq
tr cut sort uniq head tail

Sandbox, not a desert

- A **curated list** beats the kitchen sink
- Most agents don't need a whole Linux box - they need ~5 tools used well
- "Analyse BI data to find efficiencies" sounds broad - the actual surface is tiny
- Full sandboxes solve a problem most embedded agents don't have

What would a better sandbox for light tasks look like?

Introducing Pydantic Monty

- A **minimal Python interpreter, written in Rust**, for code written by AI
- Safe by default: no filesystem, env, or network unless you grant it
- Pass in your own functions - the model calls them as ordinary Python
- **Type-checks** code (via `ty`) before it runs
- **Snapshottable** - pause at an external call, resume later, anywhere
- Monty code is run in a pool of subprocesses, wire protocol for communicating with the main process

Tradeoff: Monty will not run your existing Python app - it's a Python *subset*.

- **uv** add pydantic-monty
- **npm** i @pydantic/monty
- **cargo** add monty

The advantages

STARTUP LATENCY

5 μ s

MONTY

VS.

~~1s~~

FULL SANDBOX

SNAPSHOT SIZE

1KB

MONTY

VS.

~~1GB~~

FULL SANDBOX

COST PER RUN

\$0

MONTY

VS.

~~\$?~~

FULL SANDBOX

Hack Monty

- Public bounty at hackmonty.com
- Scape the sandbox, read the secret and we'll pay **\$10,000**
- Round 2 launched a month ago
- No sandbox vulnerability found

Hack Monty

Welcome. This is a honeypot. The server behind it executes whatever Python you POST to /run/ inside [pydantic monty](#), our language-level sandbox.

There is a secret on this machine. Your job is to find it — by escaping the sandbox. If you do, we'll pay you \$5,000 USD.

[Blog post about this.](#)

[Also look at the API docs.](#) (or [Redos](#))

[Grab the hackmonty.py CLI](#) to run code here from your terminal.

[View the traces](#) in Pydantic Logfire to see how your code is executed (and everyone else's!).



Demo



Coming soon...

We also have a plan to solve agents for heavy tasks:

```
async with logfire.sandbox(provider='modal', api_key=LOGFIRE_TOKEN) as session:  
    result = await session.run_feed(code, external_functions={'my_local_function': ...})
```

Thank you



pydantic.dev/contact

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